

Correction notice (2026-05-25): The original report (v1) misread the MCQ bubble panel, recording 5/12 instead of the correct 12/12, and incorrectly scored two Section B items and one Section C item. After re-examination of higher-quality screenshots with Atif, the marks have been agreed and locked: MCQ 12/12, Section B 26/33, Section C 19/20, total 57/65 (87.7%, Grade A). This is the corrected version (v2).

Mathematics — Exam 04 (Ch 4: Factorization & Algebraic Manipulation)

Student: Seemab | **Class:** SSC-I (Grade 9) | **Date:** 25 May 2026 | **Board:** FBISE / NBF | **Total Marks:** 65 | **Version:** 2 (Corrected)

TOTAL SCORE 57/65 Percent: 87.7%	SECTION A (MCQ) 12/12 100%	SECTION B (SHORT) 26/33 78.8%	SECTION C (LONG) 19/20 95.0%	GRADE A 85–89%
MCQs 12 out of 12	Short Qs 26 out of 33	Long Qs 19 out of 20	Total 57 out of 65	

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

All 12 answers verified against the answer key from higher-quality screenshot re-examination. Every bubble matches the correct answer.

#	Topic	Seemab's Answer	Correct Answer	Result
1	Factorize $ka + kb + kc$	B — $k(a + b + c)$	B — $k(a + b + c)$	✓ 1/1
2	Factorized form of $a^2 - b^2$	C — $(a + b)(a - b)$	C — $(a + b)(a - b)$	✓ 1/1
3	Factorized form of $a^4 + a^2b^2 + b^4$	B — $(a^2 + ab + b^2)(a^2 - ab + b^2)$	B — $(a^2 + ab + b^2)(a^2 - ab + b^2)$	✓ 1/1
4	Factors of $x^2 - 9x + 8$	A — $(x - 1)(x - 8)$	A — $(x - 1)(x - 8)$	✓ 1/1
5	$a^3 + b^3$ equals	C — $(a + b)(a^2 - ab + b^2)$	C — $(a + b)(a^2 - ab + b^2)$	✓ 1/1
6	$a^3 + 3a^2b + 3ab^2 + b^3$ equals	B — $(a + b)^3$	B — $(a + b)^3$	✓ 1/1
7	LCM: highest power of which factors?	C — Highest power of every factor	C — Highest power of every factor	✓ 1/1
8	If $HCF = H$, then LCM equals	B — $P \times Q \div H$	B — $P \times Q \div H$	✓ 1/1
9	Algebraic fraction P/Q condition	C — $Q \neq 0$	C — $Q \neq 0$	✓ 1/1
10	Square root of $4a^2 + 12ab + 9b^2$	A — $\pm(2a + 3b)$	A — $\pm(2a + 3b)$	✓ 1/1
11	Adding fractions with different denominators	B — LCM of denominators	B — LCM of denominators	✓ 1/1
12	Factorization is opposite of which process?	C — Multiplication	C — Multiplication	✓ 1/1

Section A Total: 12/12 (100%) — Perfect score. MCQ streak continues across all 12 exams.

Section B — Short Questions (11 × 3 = 33 Marks)

Q.2(i) — Factorize: $2x^2y^3 - 6x^2y^2 + 2xy^3$ [Main question]

3/3

HCF extracted correctly as $2xy^2$, giving $2xy^2(xy - 3x + y)$. Answer matches the key exactly. All three method marks awarded.

HCF = $2xy^2$ identified ✓ [1]

Factored form correct ✓ [1]

Final answer ✓ [1]

Q.2(ii)(or) — Factorize: $1 - x^2 - 2xy - y^2$ [OR chosen]**2/3**

The correct approach: rewrite as $1 - (x^2 + 2xy + y^2) = 1 - (x + y)^2 = [1 + (x+y)][1 - (x+y)] = (1 + x + y)(1 - x - y)$.

Working attempts the grouping correctly up to a point — perfect-square recognition is attempted, but the final factored form $(1 + x + y)(1 - x - y)$ is not reached. The working stalls at an intermediate step along the lines of $-(x^2 - y^2)$ or a similar incomplete regrouping. Two of the three steps (grouping attempt and recognizing $(x+y)^2$ structure) are present; the application of difference-of-squares to $1 - (x+y)^2$ is missing.

Grouping attempt: $-(x^2 + 2xy + y^2) \checkmark$ [1] Perfect square $(x+y)^2$ recognized $\checkmark$ [1]

Final form $(1+x+y)(1-x-y)$ not reached [0]

Q.2(iii) — Factorize: $x^4 + 4m^4$ (Sophie Germain identity) [Main question]**2/3**

The correct method (Sophie Germain identity): add and subtract $4x^2m^2$, so $x^4 + 4m^4 = (x^2 + 2m^2)^2 - (2xm)^2 = (x^2 + 2m^2 + 2xm)(x^2 + 2m^2 - 2xm)$.

Seemab recognized the sum-of-squares structure and set up a perfect-square trinomial addition. However, the final boxed answer is written as $(x^2 + 2m^2)^2$, which expands to $x^4 + 4x^2m^2 + 4m^4$ — not the same as $x^4 + 4m^4$. The Sophie Germain identity was not completed: the step of then applying difference-of-squares to $(x^2 + 2m^2)^2 - (2xm)^2$ is absent.

Strategy recognized and add/subtract $4x^2m^2$ attempted [1 mark]. Perfect-square trinomial setup present [1 mark]. Final answer $(x^2 + 2m^2)^2$ is incorrect — it collapses the two factors back into a square rather than factoring further [0 marks for the final step].

Sum-of-squares recognized; add/subtract $4x^2m^2 \checkmark$ [1] Perfect square trinomial $(x^2 + 2m^2)^2 - (2xm)^2$ set up $\checkmark$ [1]

Did not apply difference-of-squares; final answer incorrect [0]

Q.2(iv) — Factorize: $y^2 - 7y + 12$ [Main question]**3/3**

Middle term split correctly: $-7y = -4y - 3y$. Grouped as $y(y-4) - 3(y-4) = (y-4)(y-3)$. Matches the key perfectly.

Factor pair $(-4, -3)$ identified $\checkmark$ [1] Grouping $\checkmark$ [1] Final factors $(y-4)(y-3) \checkmark$ [1]

Q.2(v) — Factorize: $x^3 + 27y^3$ (sum of cubes) [Main question]**2/3**

Correct: $a = x$, $b = 3y$. Applying $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ gives $(x + 3y)(x^2 - 3xy + 9y^2)$.

Seemab's working shows $(x + 3y)(x^2 - 6xy + 9y^2)$. The middle term is written as $-6xy$ instead of $-3xy$.

The first and third terms of the trinomial (x^2 and $9y^2$) are correct, but the middle term $-ab = -(x)(3y) = -3xy$ was doubled to $-6xy$. This is a sum-of-cubes formula application error: the trinomial factor is $(a^2 - ab + b^2)$ where $ab = (x)(3y) = 3xy$, not $6xy$. It appears the perfect-square formula middle term ($2ab$) was used instead of the sum-of-cubes formula middle term (ab). Lost 1 mark on the final answer.

Identified $a = x$, $b = 3y \checkmark$ [1] Sum-of-cubes formula applied (structure correct) $\checkmark$ [1]

Middle term $-6xy$ instead of $-3xy$: wrong final answer [0]

Q.2(vi)(or) — Factorize: $4x^2(2x - 15) - 25(5 - 6x)$ [OR chosen]**1/3**

The correct expansion: $4x^2(2x - 15) - 25(5 - 6x) = 8x^3 - 60x^2 - 125 + 150x$. Rearranged: $8x^3 - 60x^2 + 150x - 125 = (2x)^3 - 3(2x)^2(5) + 3(2x)(5^2) - 5^3 = (2x - 5)^3$.

Sign error in the expansion of $-25(5 - 6x)$: the term $-25 \times (-6x)$ should give $+150x$, but was written as $-150x$. This produces $8x^3 - 60x^2 - 150x - 125$, which does not factor as a binomial cube. The recognition of the $(2x-5)^3$ pattern was never reached because the polynomial was wrong from line two onward. The refactoring attempt is therefore based on the wrong polynomial and is meaningless. Lost 2 marks: the expansion is partially correct in the $8x^3$ and $-60x^2$ terms, hence 1 mark awarded for the partial expansion.

$8x^3 - 60x^2$ terms correct $\checkmark$ [1] Sign error on $150x$ term: $-150x$ instead of $+150x$ [0]

$(2x-5)^3$ not reached; incomplete [0]

Q.2(vii) — HCF of $(x + y)^2$ and $x^2 - y^2$ [Main question]**3/3**

Factorized $(x+y)^2 = (x+y)(x+y)$ and $x^2 - y^2 = (x+y)(x-y)$. HCF = $(x+y)$ using the lowest-power rule. Correct and clearly presented.

$(x+y)^2$ factored ✓ [1] **$x^2 - y^2$ factored ✓ [1]** **HCF = $(x+y)$ ✓ [1]**

Q.2(viii)(or) — HCF of $x^2 - 49$ and $x^2 - 4x - 21$ [OR chosen]**3/3**

(1) $x^2 - 49 = (x + 7)(x - 7)$. (2) $x^2 - 4x - 21 = x^2 - 7x + 3x - 21 = x(x-7) + 3(x-7) = (x+3)(x-7)$. HCF = $x - 7$. Both factorizations correct, HCF clearly stated.

$x^2 - 49 = (x+7)(x-7)$ ✓ [1] **$x^2 - 4x - 21 = (x+3)(x-7)$ ✓ [1]** **HCF = $x-7$ ✓ [1]**

Q.2(ix)(or) — LCM of $x^2 + x - 6$, $x^2 + x - 2$, $x^2 - 4x + 3$ [OR chosen]**1/3**

The correct factorizations: $x^2 + x - 6 = (x+3)(x-2)$; $x^2 + x - 2 = (x+2)(x-1)$; $x^2 - 4x + 3 = (x-3)(x-1)$. LCM = $(x+3)(x+2)(x-1)(x-2)(x-3)$ — all five distinct linear factors each to power one.

The three trinomials are not factored correctly in the working. The splits shown are incomplete: for x^2+x-6 , the pair (3, -2) satisfying product = -6 and sum = +1 was not identified properly. For x^2+x-2 , the pair (2, -1) similarly not found. For x^2-4x+3 , the pair (-3, -1) not identified. The LCM line written is meaningless (appears to add coefficients rather than multiply factors). The approach of factorizing each trinomial first is present (1 mark for awareness of method), but none of the three factorizations are completed correctly.

Factorization approach recognized ✓ [1] **All three trinomials unfactored or incorrectly split [0]**
LCM not correctly derived [0]

Q.2(x)(or) — Simplify: $(r^2 - s^2)/r \div (r - s)/s$ [OR chosen]**3/3**

Inverted divisor correctly: $(r^2-s^2)/r \times s/(r-s)$. Factorized $r^2-s^2 = (r+s)(r-s)$. Cancelled $(r-s)$. Result: $s(r+s)/r = (sr + s^2)/r$. This is equivalent to the key's answer. Methodology clean and complete. Handwriting benefit of doubt on final form accepted.

Invert divisor ✓ [1] **r^2-s^2 factored and $(r-s)$ cancelled ✓ [1]** **Final answer $(rs + s^2)/r$ ✓ [1]**

Q.2(xi) — Find square root of $4a^2 + 12ab + 9b^2$ by factorization [Main question]**3/3**

$4a^2 + 12ab + 9b^2 = (2a)^2 + 2(2a)(3b) + (3b)^2 = (2a + 3b)^2$. Square root = $\pm(2a + 3b)$. Stray superscript notation on the final line accepted — the substance is correct.

Perfect square form identified ✓ [1] **Complete square factored ✓ [1]** **Square root $\pm(2a + 3b)$ stated ✓ [1]**

Section B Total: 26/33. Deductions: Q.2(ii)(or) -1 (final step not reached), Q.2(iii) -1 (Sophie Germain identity incomplete), Q.2(v) -1 (wrong middle term in sum-of-cubes: $-6xy$ instead of $-3xy$), Q.2(vi)(or) -2 (sign error on $150x$, $(2x-5)^3$ not reached), Q.2(ix)(or) -2 (trinomials unfactored, LCM wrong).

Section C — Long Questions (4 × 5 = 20 Marks)**Q.3(i) — Factorize: $(x^2 + 3x - 4)(x^2 + 3x + 5) + 8$ [Main question]****5/5**

Let $a = x^2 + 3x$. Expression becomes $(a - 4)(a + 5) + 8$.

Expanded: $a^2 + 5a - 4a - 20 + 8 = a^2 + a - 12$.

Factorized: $a^2 + 4a - 3a - 12 = a(a + 4) - 3(a + 4) = (a - 3)(a + 4)$.

Substituted back: $(x^2 + 3x - 3)(x^2 + 3x + 4)$.

Substitution strategy perfectly executed. All five steps present and correct. Final answer $(x^2 + 3x - 3)(x^2 + 3x + 4)$ matches the key exactly. This is the hardest question-type in Ch4 and it was handled with no hesitation.

Substitution $a = x^2+3x$ ✓ [1] **Expansion a^2+a-12 ✓ [1]** **Factorize a^2+a-12 ✓ [1]** **Back-substitution ✓ [1]**
Final answer ✓ [1]

Q.3(ii) — HCF of $P = 2m^2 - 2mn$, $Q = 4m^4 - 4m^2n^2$, $R = 2m^3 - 4m^2n + 2mn^2$ [Main question]
5/5

$$P = 2m^2 - 2mn = 2m(m - n)$$

$$Q = 4m^4 - 4m^2n^2 = 4m^2(m^2 - n^2) = 4m^2(m - n)(m + n) = 2^2 \cdot m^2 \cdot (m - n) \cdot (m + n)$$

$$R = 2m^3 - 4m^2n + 2mn^2 = 2m(m^2 - 2mn + n^2) = 2m(m - n)^2$$

$$\text{HCF (lowest power of each common factor): } 2^1 \cdot m^1 \cdot (m - n)^1 = 2m(m - n)$$

All three factorizations executed correctly. Lowest-power rule applied precisely. HCF = $2m(m - n)$ is confirmed correct.

$$P = 2m(m - n) \checkmark [1]$$

$$Q = 4m^2(m - n)(m + n) \checkmark [1]$$

$$R = 2m(m - n)^2 \checkmark [1]$$

Common factors identified $\checkmark [1]$

$$\text{HCF} = 2m(m - n) \checkmark [1]$$

Q.3(iii) — LCM of $2x^2 - 12xy + 16y^2$, $x^2 - 6xy + 8y^2$, $3x^2 - 12y^2$; verify $\text{LCM} \times \text{HCF} = P \times Q$ [Main question]
4/5

$$(1) 2x^2 - 12xy + 16y^2 = 2(x^2 - 6xy + 8y^2) = 2(x - 4y)(x - 2y) \checkmark$$

$$(2) x^2 - 6xy + 8y^2 = (x - 4y)(x - 2y) \checkmark$$

$$(3) 3x^2 - 12y^2 = 3(x^2 - 4y^2) = 3(x + 2y)(x - 2y) \checkmark$$

$$\text{LCM} = 6(x - 4y)(x - 2y)(x + 2y) \checkmark$$

All three factorizations correct. LCM correctly computed as $6(x - 4y)(x - 2y)(x + 2y)$.

The verification working ($\text{LCM} \times \text{HCF} = P \times Q$, typically done for the first two polynomials P and Q) is present but the intermediate steps are algebraically messy — there are expanding errors visible in the middle of the calculation that produce exponents that do not match. The verification does ultimately land on the correct equality $2(x - 4y)^2(x - 2y)^2 = 2(x - 4y)^2(x - 2y)^2$ at the final line, but the route there is not clean enough to warrant full marks on the verification portion. One mark deducted for the messy intermediate verification steps.

$$\text{Polynomial (1) factored } \checkmark [1]$$

$$\text{Polynomial (2) \& (3) factored } \checkmark [1]$$

$$\text{LCM} = 6(x - 4y)(x - 2y)(x + 2y) \checkmark [1]$$

$$\text{Verification approach correct, final line right } \checkmark [1]$$

Messy intermediate steps in verification $[-1]$

Q.3(iv) — Simplify: $-2x/(x+3) + 3/(3-x) - (8x-12)/(x^2-9)$ [Main question]
5/5

$$\text{Step 1: Denominators — } (x+3), (3-x) = -(x-3), (x^2-9) = (x+3)(x-3)$$

$$\text{Step 2: Rewrote } 3/(3-x) \text{ as } -3/(x-3), \text{ handling the sign change cleanly}$$

$$\text{Step 3: LCM of denominators} = (x+3)(x-3)$$

$$\text{Step 4: Combined numerator} = -2x(x-3) - 3(x+3) - (8x-12) = -2x^2+6x - 3x-9 - 8x+12 = -2x^2 - 5x + 3$$

$$\text{Step 5: } -2x^2 - 5x + 3 = -(2x^2+5x-3) = -(2x-1)(x+3). \text{ Cancelled } (x+3). \text{ Result} = -(2x-1)/(x-3) = (1-2x)/(x-3)$$

Excellent handling of the sign change on $3/(3-x)$. Numerator expansion and algebra clean. Factorization $-(2x-1)(x+3)$ correct. Final answer $(1-2x)/(x-3)$ matches the key precisely.

$$\text{Denominators factored } \checkmark [1]$$

$$\text{Sign change on } 3/(3-x) \text{ handled } \checkmark [1]$$

$$\text{LCM and combined numerator } \checkmark [1]$$

$$\text{Numerator factorized } \checkmark [1]$$

$$\text{Cancellation and final answer } \checkmark [1]$$

Section C Total: 19/20 — deduction on Q.3(iii): messy intermediate steps in the verification portion (–1 mark).

Final Mark Tally (Corrected)

Section	Marks Awarded	Out Of	Percentage
Section A (MCQs)	12	12	100%
Section B (Short Questions)	26	33	78.8%
Section C (Long Questions)	19	20	95.0%
TOTAL	57	65	87.7%

Performance Analysis

Strengths in this exam

- Perfect MCQ score (12/12) — all definitional and identity questions answered correctly
- Substitution factorization: Q.3(i) 5/5, all five steps present and clean
- HCF of three polynomials: Q.3(ii) 5/5, lowest-power rule applied precisely
- Algebraic fractions with sign-change denominator: Q.3(iv) 5/5, $3/(3-x)$ handled correctly
- HCF by factorization for simpler cases: Q.2(vii) and Q.2(viii)(or) both full marks
- Square root by factorization: Q.2(xi) 3/3
- Division of algebraic fractions: Q.2(x)(or) 3/3
- HCF extraction for common factor: Q.2(i) 3/3

Areas for improvement

- **Sum-of-cubes middle term (Q.2(v)):** $a^3+b^3 = (a+b)(a^2 - ab + b^2)$. The middle term is $-ab$, not $-2ab$. Seemab used $-2ab$ (the perfect-square coefficient). Drill: explicitly write out ab and confirm it is not doubled.
- **Sophie Germain identity completion (Q.2(iii)):** After setting up $(x^2+2m^2)^2 - (2xm)^2$, the second step of applying difference-of-squares was skipped. The answer stopped one step too early.
- **Grouping with difference-of-squares as final step (Q.2(ii)(or)):** $1 - (x+y)^2$ was recognized but not factored into $(1+(x+y))(1-(x+y))$.
- **Sign discipline in binomial expansion (Q.2(vi)(or)):** $-25 \times (-6x) = +150x$ was written as $-150x$. This single sign error collapsed the entire question.
- **LCM of three trinomials (Q.2(ix)(or)):** None of the three trinomials were fully factored. Need more practice with splitting the middle term when the product is negative.
- **Verification step neatness (Q.3(iii)):** The conclusion was correct but intermediate algebraic steps were unclear. Work should be presented in a clean column with each step on a separate line.

Comparison: Math Exam History

Exam	Date	Chapters	MCQ	Section B	Section C	Total	%
Math exam-01-ch1-2	18 Apr 2026	Ch 1-2	12/12	33/33	20/20	65/65	100%
Math exam-03-ch4-5-7	22 Apr 2026	Ch 4, 5, 7	12/12	28.5/33	20/20	60.5/65	93.1%
Math exam-04-ch4 (this)	25 May 2026	Ch 4	12/12	26/33	19/20	57/65	87.7%

MCQ mastery maintained at 12/12 across all three Math exams. Section B dipped (26/33 vs 28.5/33 last time) due to formula errors on sum-of-cubes and Sophie Germain, plus the LCM trinomials question. Section C is strong at 19/20. The overall drop from 93.1% to 87.7% is entirely in Section B formula application, not in concept understanding.

Priority Revision Targets

- Sum-of-cubes middle term:** $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$. The middle term is exactly $-ab$ (not $-2ab$, not $+ab$).
Write this identity ten times, then verify by expanding.
- Sophie Germain identity — two steps:** $x^4 + 4m^4$. Step 1: add and subtract $4x^2m^2$ to form $(x^2+2m^2)^2 - (2xm)^2$.
Step 2: apply difference-of-squares. Do not stop after Step 1.
- Difference-of-squares applied to non-numeric squares:** $1 - (x+y)^2$ is $a^2 - b^2$ where $a=1$, $b=(x+y)$. Practice this pattern specifically: $[1+(x+y)][1-(x+y)]$.
- Sign discipline in bracket expansion:** When expanding $-k(a - b)$, the second term is $+kb$. Slow down and explicitly write the sign before each term as a separate step.
- LCM of trinomials — split-middle-term drill:** For $x^2 + bx + c$ with c negative, list factor pairs of c (one positive, one negative) until the pair summing to b is found. This is the most common error pattern in this exam.
- Verification step layout:** Write $\text{LCM} \times \text{HCF}$ on the left, $P \times Q$ on the right, expand both fully in a column format before comparing. Do not try to shortcut across the page.

